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Hiring Manager
Twitch
San Francisco, CA

Dear Hiring Manager,

My name is Ajay Venkatesh. I'm finishing my M.S. in Computer Engineering at NYU, with production software experience at PwC in Bengaluru, a summer SDE internship at Amazon, and ongoing on-campus work at NYU's Novel AI Technologies. I'm writing about the Software Engineer, ML Products role on the discovery team.

The strongest match is the backend and AWS infrastructure work I owned at Amazon. I provisioned a **serverless Infrastructure-as-Code** stack on **AWS CDK** using **API Gateway**, **Lambda**, **DynamoDB**, **S3**, and **IAM**, deployed through multi-stage and multi-region pipelines, and built backend services in **Java** using **Coral**, **Smithy**, **Dagger**, and **CBOR**. I also worked directly with **RAG** and **Model Context Protocol** retrieval pipelines on **ECS Fargate**, **SQS**, and **SNS**, decoupling distributed message processing across a high-throughput triage system. That gives me real, recent muscle in exactly the AWS surface this role lives on.

In parallel, at Campus Mesh I worked on the systems behind a **recommendation engine** built on collaborative filtering and behavioral analytics, plus **LLM-based retrieval pipelines (RAG)** over an in-app AI assistant, with **embedding-based semantic search** and structured indexing for contextual accuracy. I modeled high-throughput chat in **DynamoDB** with partition and sort keys, and designed schemas to keep traffic balanced. Recommendations and search are not abstract problems for me; they are the work I've actually shipped.

On overlap with your stack: **Python**, **Java**, **TypeScript**, plus **AWS** (**S3, ECS, Lambda, DynamoDB, SageMaker, SQS, SNS, API Gateway, Athena**), **distributed systems**, **API design**, and **data modeling** from production work. Graduate coursework covered **Machine Learning**, **Deep Learning**, and **Big Data**, so I'm comfortable working alongside applied scientists to take models from notebook to production. **Go** is the one stack item I haven't shipped in production, but I've picked up new backend languages quickly before, including Java at Amazon and Node.js at Campus Mesh.

What pulls me toward Twitch is the surface itself. Discovery on a live streaming platform is one of the most interesting recommendation problems in the industry: it is real-time, the inventory is constantly changing, the signal is noisy, and the consequences for viewers and creators are immediate. Working on the systems that decide who finds whom, at the scale Twitch operates, is the kind of long-tail engineering work that compounds.

I'd welcome the chance to discuss further. Thanks for considering my application.

Sincerely,
Ajay Venkatesh